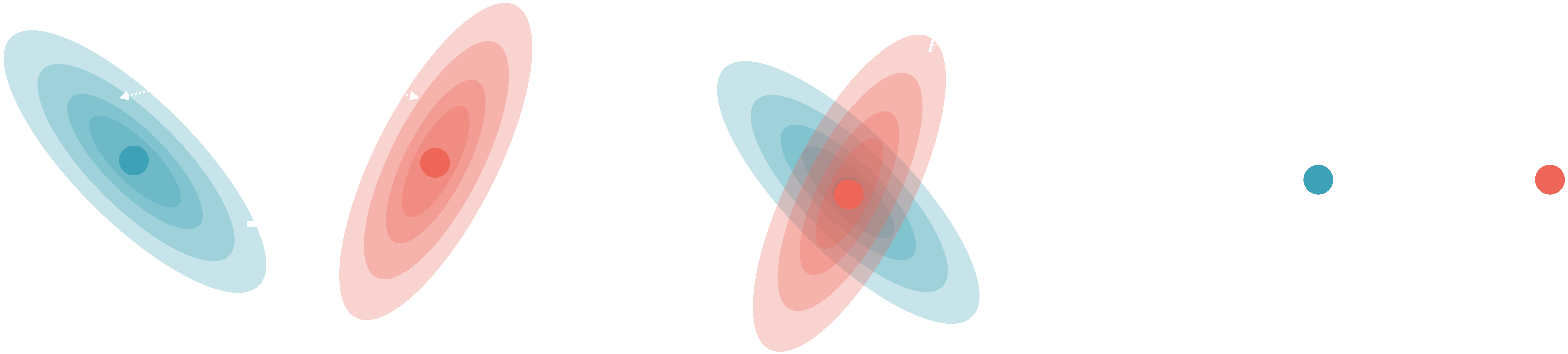


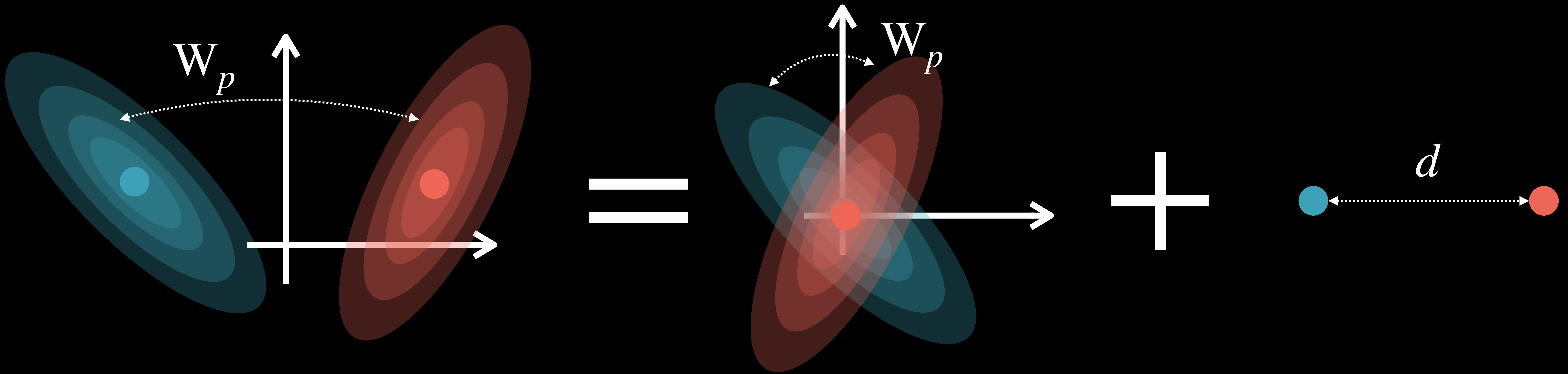
W_2 factors out translations

Consider the translation operator $T_\tau : x \mapsto x - \tau$. One has:

$$\mathrm{W}_2(T_{\tau\sharp}\alpha, T_{\tau'\sharp}\beta)^2 \;=\; \mathrm{W}_2(\alpha,\beta)^2 - 2\langle \tau - \tau', \mu_\alpha - \mu_\beta\rangle + \|\tau - \tau'\|^2$$

In particular, this implies

$$\mathrm{W}_2(\alpha,\beta)^2 \;=\; \mathrm{W}_2(\tilde{\alpha},\tilde{\beta})^2 + \|\mu_\alpha - \mu_\beta\|_2^2$$



Where $\tilde{\alpha}, \tilde{\beta}$ are centered (mean 0) translations of α, β

Section 2: Metric Properties